CLIENT WALKTHROUGH

# Sensors & Simulation

A practical, step-by-step guide for testing sensor management and simulated machine readings in the PredictorMT user dashboard.

TEST FLOW

| 1 Prepare | 2 Register | 3 Simulate | 4 Review | 5 Finish |
|---|---|---|---|---|
| Company machine | Assign a sensor | Generate readings | Charts and alerts | Stop the run |

## Before you begin

- Sign in with a user account that belongs to the company you are testing.
- Have at least one machine registered under that company. Create one first if needed.
- For automatic readings, the server scheduler must be running. New readings are generated on its schedule (once per minute).
- Use a simulation scenario to test generated data. Simulation does not connect to or verify real sensor hardware.

**IMPORTANT: simulation versus a real device**

Simulation creates sample readings inside the application and may create test alerts or maintenance tasks. It does not prove that a physical sensor is installed, connected, calibrated, or transmitting.

## In this guide

- Page 2: add and verify a sensor.
- Page 3: run a simulation and review its readings.
- Page 4: test checklist and common troubleshooting.

# 1. Register a sensor

A sensor record identifies a device and links it to a machine in your company. The machine assignment is required.

**1**

### Open Sensor Management

In the dashboard sidebar, choose Sensor Management, then select the create/add sensor action.

**2**

### Enter the device details

Provide a unique Device ID and a Sensor Type (for example, temperature or multi-parameter).

**3**

### Assign the company machine

Choose the correct machine from the Machine list. Only machines in your company should be available.

**4**

### Complete optional fields and save

Firmware version, battery level, and last-seen time are optional. Choose a status, then select Create Sensor.

## Confirm the sensor was saved

- The sensor appears in the sensor list with the expected device ID, type, machine, and status.
- Open its details, then edit and save a harmless field change to verify update access.
- A sensor record alone does not produce readings. Readings arrive from a real ingestion connection or from simulation.

**Expected result**

The sensor is visible to your company and assigned to the selected machine. Another company must not be able to view or change it.

# 2. Run a machine simulation

Simulation generates sample sensor readings for a selected company machine. It is the recommended way to test dashboard charts and alerts without a physical device.

**1**

### Open Simulation and choose a machine

Use the dashboard sidebar. Select the company machine you want to test.

**2**

### Choose a scenario and speed

Start with Normal operation for a baseline. A fault scenario can be used to check fault handling.

**3**

### Review thresholds and start

Keep the default thresholds for a basic test, then select Start. Optional calibration adjusts simulated temperature.

**4**

### Wait for and review readings

Readings are generated by the server scheduler (once per minute). Check the charts, latest readings, and simulation status.

### What the controls do

- Pause stops new readings while preserving the run; Start can resume when the machine is ready.
- Stop ends the run. Historical playback loads saved readings for a selected time range.
- The speed setting affects generated simulation values; it does not make the scheduler run more often.

**Fault-scenario test note**

Starting most fault scenarios records a scenario alert and a maintenance task. Readings that cross configured thresholds can also create alerts. Use a test machine and review/clean up test records according to your process.

# 3. Test checklist & results

Run these checks in order and record the outcome. Use a company test machine and avoid deleting production records.

### 1. Machine prerequisite

PASS / FAIL

A machine belonging to the test company is listed in Simulation.

### 2. Sensor creation

PASS / FAIL

A sensor saves with a unique ID, type, required machine, and selected status.

### 3. Company access

PASS / FAIL

The company user can view and update its sensor; a different company cannot.

### 4. Normal simulation

PASS / FAIL

Start Normal operation; status changes and readings appear after a scheduler tick.

### 5. Charts and history

PASS / FAIL

Charts/table update; historical playback can load saved readings for a date range.

### 6. Pause and stop

PASS / FAIL

Pause prevents new readings; Stop ends the run.

### 7. Fault scenario (optional)

PASS / FAIL

A test fault scenario produces the expected simulation data and scenario event.

## Troubleshooting

- No readings: verify the scheduler is enabled and running; allow up to a minute for the next scheduled tick.
- Machine missing: confirm it belongs to the signed-in user company.
- Start rejected: check that machine state is Ready and correct maintenance, fault, or emergency-stop state first.
- No threshold alert: confirm values actually crossed the configured threshold; normal sample data may not cross it.